

Required practical: Resistance

Answers

Year 10 GCSE Physics, AQA Trilogy RP 15; Physics RP 3

Name _____

Class _____ Date _____

Plan. Measure. Explain.

Work in pairs. Read the whole method before you build anything.

Online version with hints: panphy.app/gcsephy/year10phy/unit01/practical/resistance/

1 Aim

Use circuit diagrams to set up circuits that investigate the factors affecting resistance: the length of a wire at constant temperature, and combinations of resistors in series and parallel.

$$R = V \div I$$

R in ohms (Ω), V in volts (V), I in amperes (A)

2 Apparatus

- Low-voltage d.c. power supply or battery (about 2 V)
- Ammeter (0–2 A)
- Voltmeter (0–5 V)
- Switch and connecting leads
- Two crocodile clips
- Metre ruler and sticky tape
- About 1.1 m of thin constantan or nichrome wire
- Two identical resistors, for example 10 Ω (Part B)

3 Safety

- The thin wire can get hot enough to burn. Do not touch it while the current is on.
- Switch off between readings. This also stops the wire heating up.
- Keep the p.d. low. Never connect the wire straight across the supply without the ammeter in the circuit.

Before you start

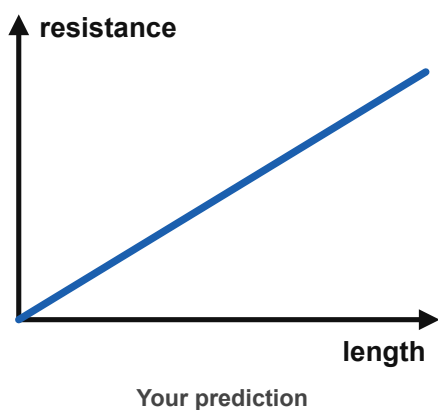
Learn the key words, then make a prediction before you build anything.

4 Key words

Independent variable	The variable you change.
Dependent variable	The variable you measure.
Control variable	A variable you keep the same, so the test is fair.
Resolution	The smallest change a meter can show, for example 0.01 A.
Anomalous result	A result that does not fit the pattern of the others.
Line of best fit	One straight line (or smooth curve) drawn as close as possible to all the points.

5 Predict

Sketch how you think the resistance of the wire will change as its length increases.



Expected: a straight line through the origin. Resistance is directly proportional to length.

Q1 Give a reason for your prediction.

A longer wire has more metal ions along the electrons' path, so there are more collisions. Doubling the length should double the resistance.

Part A Length of a wire

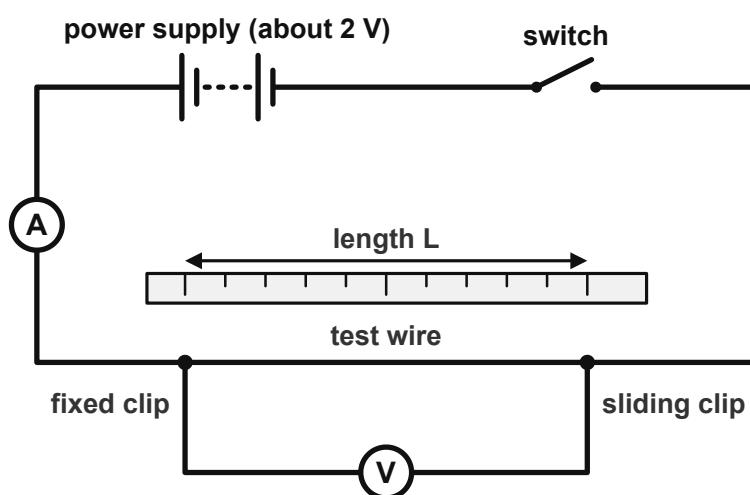
Resistance, Part A: Change the length of wire between two crocodile clips. Measure the p.d. and current each time and calculate the resistance.

1 Variables

Variable	What it is	How
Independent	Length of the wire between the clips	Move the sliding clip in 10 cm steps
Dependent	Resistance of the wire	Measure V and I, then calculate $R = V \div I$
Control	Material and thickness (diameter) of the wire	Use the same piece of wire throughout
Control	Temperature of the wire	Low p.d.; switch off between readings

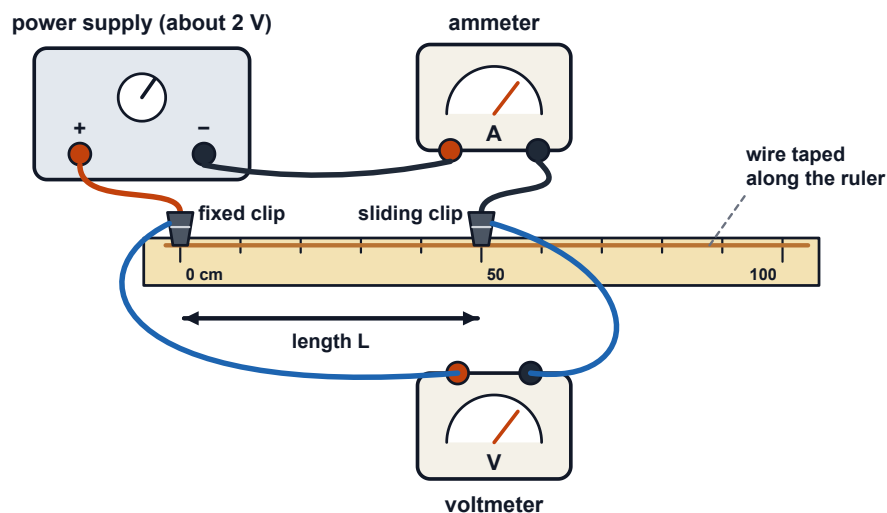
2 Method

1. Tape the wire tightly along the metre ruler so it is straight. Line up one end with the 0 cm mark.
2. Build the circuit below. Attach the fixed crocodile clip at 0 cm and the sliding clip at 10 cm.
3. Close the switch. Read the voltmeter and the ammeter quickly, then open the switch.
4. Move the sliding clip 10 cm further along the wire and repeat step 3. Carry on up to 100 cm.
5. Calculate the resistance for each length using $R = V \div I$.
6. Repeat any reading that does not fit the pattern of the others.

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The wire is taped along a metre ruler. The voltmeter is connected across the wire between the two crocodile clips.

Part A Length of a wire



On the bench: the ammeter is in series; the voltmeter leads clip on at the same points as the crocodile clips. Include a switch in the circuit.

Measure carefully. Measure each length to the same contact point on the crocodile clips, and read the ruler at eye level.

3 Results

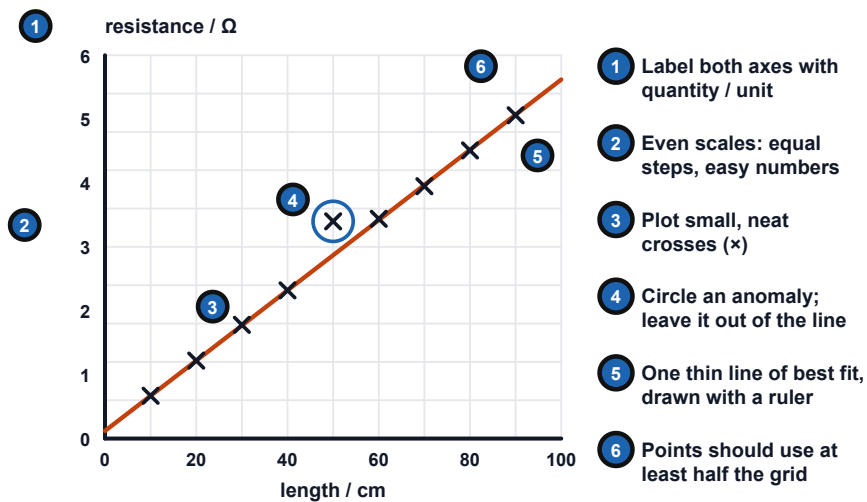
Give p.d. and current to 2 decimal places. Give resistance to 2 decimal places.

Length / cm	p.d. / V	Current / A	Resistance / Ω
10	0.72	1.07	0.67
20	1.01	0.83	1.22
30	1.19	0.67	1.78
40	1.32	0.57	2.32
50	1.41	0.49	2.88
60	1.48	0.43	3.44
70	1.54	0.39	3.95
80	1.58	0.35	4.51
90	1.62	0.32	5.06
100	1.65	0.29	5.69

Part A Length of a wire

Q2 Look at your resistance column. Describe how the resistance changes as the length of the wire increases.

As the length increases, the resistance increases. Doubling the length roughly doubles the resistance (for example, $1.22\ \Omega$ at 20 cm and $2.32\ \Omega$ at 40 cm).

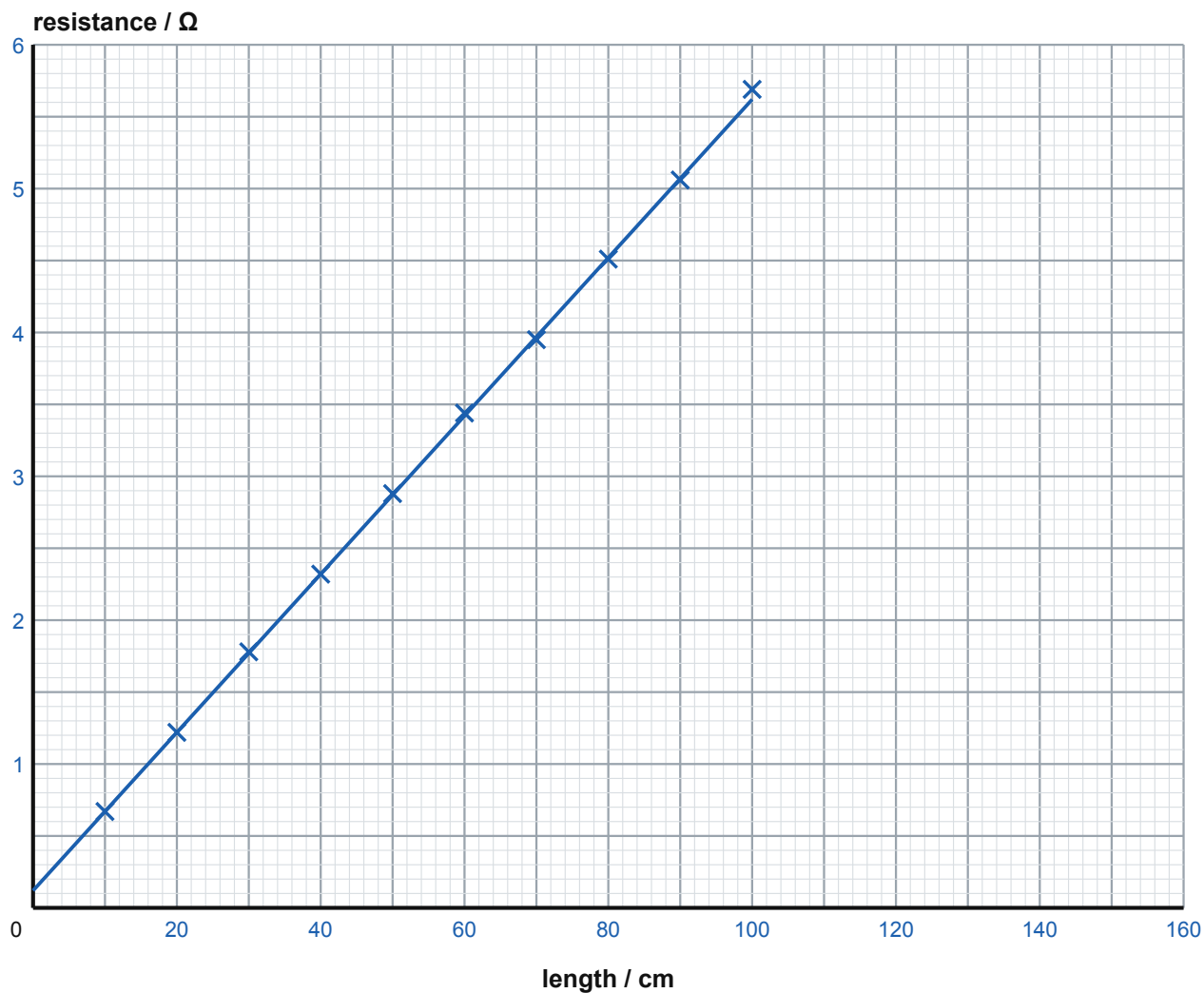


A good graph: the numbered checks are how graph marks are awarded.

Part A Length of a wire

4 Plot your graph

Plot resistance (vertical axis) against length (horizontal axis). Choose scales that use at least half of the grid. Draw a straight line of best fit.



Q3 Describe your graph. What does it show about the relationship between resistance and length?

The graph is a straight line that passes through (or very close to) the origin. This shows that the resistance of the wire is directly proportional to its length.

Part A Length of a wire

Q4 Use your graph to find the resistance of 45 cm of this wire. Show on your graph how you did this.

Read up from 45 cm to the line of best fit and across to the resistance axis: about $2.6\ \Omega$ for the example results. Use your own line; your value depends on your wire.

Q5 The line of best fit may cross the resistance axis slightly above zero. Suggest why.

The crocodile clips and their contacts with the wire have a small resistance of their own, so every reading is slightly too high by about the same amount. This is a systematic error: it moves the whole line up but does not change its gradient.

Q6 Explain why the circuit should be switched off between readings.

The current heats the wire. A hotter wire has a greater resistance, because its ions vibrate more and electrons collide with them more often. Temperature is a control variable, so the wire must stay cool for the test to be fair.

Q7 Explain, in terms of electrons, why a longer wire has a greater resistance.

Electrons collide with the ions in the metal as they move through the wire. In a longer wire there are more ions along the path, so there are more collisions and the electrons are opposed more. The resistance is greater.

Part A Length of a wire

Q8 A 30 cm length of the wire had a p.d. of 1.19 V across it and a current of 0.67 A through it. Calculate its resistance.

$R = V \div I = 1.19 \div 0.67 = 1.78 \, \Omega$ (1.8 Ω to 2 significant figures).

Q9 The ammeter shows readings such as 0.67 A. What is the resolution of the ammeter?

0.01 A. The display shows two decimal places, so the smallest change it can show is 0.01 A.

Q10 Look at your graph. Identify any anomalous result, or state that there are none. What should you do with an anomalous result?

An anomalous result is a point well away from the line of best fit. Circle it, leave it out when drawing the line, and repeat that reading if there is time.

Q11 Suggest one improvement to the method. Explain how it would make the results better.

Any one, with its reason: repeat each reading and calculate a mean, which reduces the effect of random errors and helps spot anomalies; use a lower p.d. or switch off for longer between readings, so the temperature of the wire stays constant; use more lengths (for example every 5 cm) to show the pattern more clearly.

Part B Resistors in series and parallel

Resistance, Part B: Measure the resistance of each resistor on its own, then of the two resistors joined in series and in parallel.

1 Variables

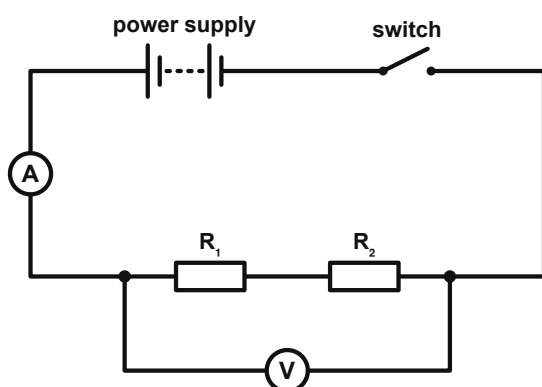
Variable	What it is	How
Independent	How the resistors are arranged	One resistor, two in series, two in parallel
Dependent	Total resistance of the arrangement	Measure V and I, then calculate $R = V \div I$
Control	The resistors and the supply setting	Use the same two resistors and supply throughout

Q12 Before you measure: predict the total resistance of R_1 and R_2 in series, and in parallel. Is each one bigger or smaller than one resistor on its own?

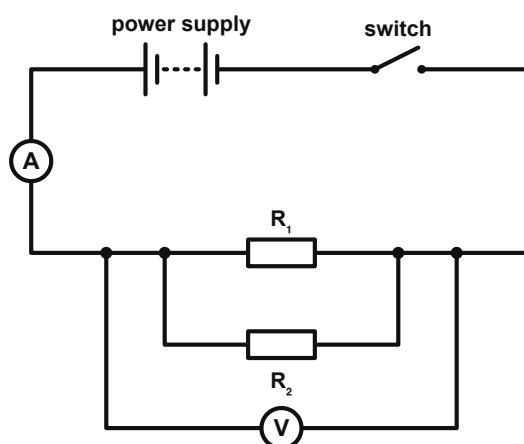
Series: bigger than either resistor (about $R_1 + R_2$, so about $20\ \Omega$ for two $10\ \Omega$ resistors). Parallel: smaller than either resistor (about $5\ \Omega$ for two $10\ \Omega$ resistors).

2 Method

1. Connect resistor R_1 on its own, with the ammeter in series and the voltmeter across the resistor. ☐
2. Close the switch. Record the p.d. and the current, then open the switch. ☐
3. Replace R_1 with R_2 and repeat step 2. ☐
4. Connect R_1 and R_2 in series, one after the other. Put the voltmeter across both resistors. Repeat step 2. ☐
5. Connect R_1 and R_2 in parallel, on separate branches between the same two junctions. Put the voltmeter across the pair. Repeat step 2. ☐
6. Calculate the resistance for each arrangement using $R = V \div I$. ☐



Series: the voltmeter goes across both resistors.



Parallel: the voltmeter goes across the pair of resistors.

Part B Resistors in series and parallel

3 Results

Give resistance to 1 decimal place.

Arrangement	p.d. / V	Current / A	Resistance / Ω
R_1 on its own	3.0	0.30	10.0
R_2 on its own	3.0	0.29	10.3
R_1 and R_2 in series	3.0	0.15	20.0
R_1 and R_2 in parallel	2.9	0.58	5.0

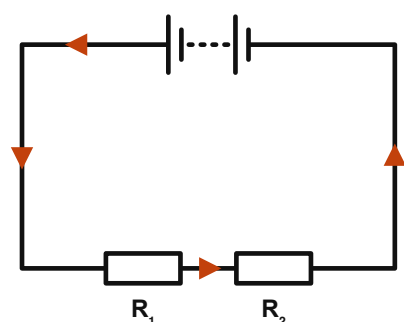
Q13 Add R_1 and R_2 from your table. Compare the answer with your measured resistance for the series arrangement.

For the example results, $R_1 + R_2 = 10.0 + 10.3 = 20.3 \Omega$ and the measured series resistance is 20.0Ω . These are approximately equal, so $R_{\text{total}} = R_1 + R_2$ for resistors in series.

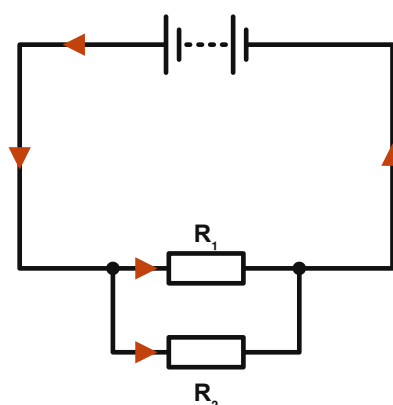
Q14 How does the resistance of the parallel arrangement compare with the resistance of R_1 and of R_2 on their own?

The parallel resistance (5.0Ω in the example) is less than the resistance of either resistor on its own. For two identical resistors it is about half the resistance of one.

Part B Resistors in series and parallel



Series: one path.
The charge goes through R_1 and then R_2 .



Parallel: two paths. More current flows
for the same p.d., so the total R is smaller.

Arrows show the conventional current. Adding a branch in parallel adds another path.

Q15 Explain why adding a resistor in parallel decreases the total resistance.

The extra resistor gives the charge another path to flow through. With the same p.d., the total current from the supply increases, so the total resistance ($R = V \div I$) is smaller.

Q16 Predict the total resistance of three of these resistors in series. What would happen to the total resistance if you kept adding more of them in parallel instead?

Three in series: about $3 \times 10 \, \Omega = 30 \, \Omega$. Adding more resistors in parallel keeps decreasing the total resistance, because each one adds another path for the current.

Q17 A $22 \, \Omega$ resistor and a $47 \, \Omega$ resistor are connected in series to a $6.0 \, \text{V}$ supply. Calculate the total resistance, then the current.

$R_{\text{total}} = 22 + 47 = 69 \, \Omega$. $I = V \div R = 6.0 \div 69 = 0.087 \, \text{A}$.

Exam practice Plan the practical

Resistance, Exam practice: A six-mark method question like the ones AQA asks about this practical. Write your answer without looking at the method.

AQA-STYLE **Q18** A student wants to investigate how the length of a wire affects its resistance. Describe a method the student could use. You may include a circuit diagram in your answer. **[6 marks]**

Marking points (level of response):

- Tape the wire to a metre ruler and connect crocodile clips to it.
- Connect an ammeter in series and a voltmeter in parallel across the length of wire between the clips.
- Measure the length between the clips with the ruler.
- Record the p.d. and current, and calculate $R = V \div I$.
- Change the length (for example, in 10 cm steps up to 100 cm) and repeat.
- Keep the temperature constant: low p.d. and switch off between readings.
- Use the same wire (material and diameter); repeat readings, find means and plot R against length.

Level 3 (5–6 marks): a clear, logical method that would give valid results. Level 2 (3–4 marks): most steps, but not fully logical or detailed. Level 1 (1–2 marks): some relevant steps.

1 Check your answer

When you have finished, tick each point your method includes.

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| ☐ A circuit: ammeter in series, voltmeter across the wire | ☐ Change the length in equal steps over a wide range |
| ☐ Measure the length between the clips with a ruler | ☐ Same wire; low p.d. and switch off between readings |
| ☐ Record V and I; calculate $R = V \div I$ | ☐ Repeat readings; plot R against length |